

Brain Connectivity Pipeline

Ergebnisbericht · Mock Data · Brodmann Atlas (82 Nodes) · 100 Subjects · 3 Sessions

1. Pipeline-Übersicht

Subjects	Sessions	Matrizen	Node-Records
100	3	300	24.600
5 Gruppen	Brodmann	82 Nodes	BCT-Metriken

Gruppen: control · intervention_alone_short · intervention_alone_long · intervention_group_short · intervention_group_long (je 20 Subjects)

2. Nodal Hub Identification

BCT-Metriken berechnet: degree, strength, betweenness centrality, clustering coefficient, local efficiency, participation coefficient, within-module z-score. Laufzeit: 55 Minuten (300 Matrizen × 82 Nodes).

3. Nodal Temporal Trajectories

2.870 Trajektorien berechnet (82 Nodes × 7 Metriken × 5 Gruppen). 574 Node-Metrik-Effekte. 463 Nodes mit |d| > 0.5 (starker Effekt).

Node	Metrik	Cohen's d
26	clustering	35.75
26	local efficiency	35.56
26	strength	29.35
34	strength	18.11
53	local efficiency	17.84

Top 5 Nodes mit stärkstem Effekt über alle Sessions.

4. Nodal Multivariate Analysis

Analyse	Gesamt	Signifikant (p<0.05)	Großer Effekt
5-Group ANOVA	574	270 (47.0%)	220 (38.3%)
Intervention vs. Control	574	283 (49.3%)	232 (40.4%)
Dauer (Short vs. Long)	574	245 (42.7%)	0 (0.0%)
Social (Alone vs. Group)	574	66 (11.5%)	0 (0.0%)
Gender Effects	—	—	nicht verfügbar
Age Correlations	—	—	age fehlt in parquet

5. SVM + Random Forest Klassifikation

Variante	RF CV	SVM CV	Bestes Modell
Intervention vs. Control	96.0 ± 2.0%	95.0 ± 1.0%	RF
Duration (Short vs. Long)	74.1 ± 4.7%	78.1 ± 7.2%	SVM
Social (Alone vs. Group)	61.1 ± 5.9%	54.0 ± 3.1%	RF
5-Groups (Time Effect)	58.0 ± 4.0%	46.0 ± 2.0%	RF

Top Features (ANOVA): clustering_slope · local_efficiency_slope · strength_slope · betweenness_slope

6. Stratifikation nach Geschlecht

Gruppe	n	CV Accuracy	Train Accuracy
Weiblich (f)	52	48.0 ± 8.0%	69.2%
Männlich (m)	48	56.2 ± 7.9%	79.2%

Age-Stratifikation nicht verfügbar (age-Spalte fehlt in node_level_metrics.parquet).

7. Pipeline-Status

Skript	Status	Anmerkung
global_basic_metrics	OK	Summary-Tabellen generiert
nodal_hub_identification	OK	24.600 Records · 55 Min
nodal_temporal_trajectories	OK	2.870 Trajektorien · 574 Effekte
nodal_multivariate_analysis	OK	6 Analysen durchgeführt
stat_svm_baseline_analysis	Warnung	Nicht genug Samples nach Remap
stat_svm_time_5groups	OK	RF CV=58% · SVM CV=46%
stat_svm_stratified_5groups	OK	Sex-Stratifikation
stat_random_forest_comparison	OK	3 Varianten: Social, Duration, Intervention